

APPENDIX I
COMPUTATION OF THE MATRIX P

Consider the symmetric matrix

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix}. \quad (1)$$

By Sylvester's criterion, P is positive definite if and only if

$$P_{11} > 0, \quad P_{11}P_{22} - P_{12}^2 > 0. \quad (2)$$

Now consider the matrix

$$A = \begin{bmatrix} A_{11} & 1 \\ A_{21} & 0 \end{bmatrix}. \quad (3)$$

Then,

$$\begin{aligned} M &:= A^\top P + PA \\ &= \begin{bmatrix} 2A_{11}P_{11} + 2A_{21}P_{12} & P_{11} + A_{11}P_{12} + A_{21}P_{22} \\ P_{11} + A_{11}P_{12} + A_{21}P_{22} & 2P_{12} \end{bmatrix}. \end{aligned} \quad (4)$$

Since M is symmetric, Sylvester's criterion implies that M is negative definite if and only if

$$M_{11} < 0, \quad \det(M) > 0. \quad (5)$$

Therefore, $A^\top P + PA < 0$ if and only if

$$A_{11}P_{11} + A_{21}P_{12} < 0, \quad (6)$$

and

$$4P_{12}(A_{11}P_{11} + A_{21}P_{12}) > (P_{11} + A_{11}P_{12} + A_{21}P_{22})^2. \quad (7)$$

Moreover, since (6) requires

$$A_{11}P_{11} + A_{21}P_{12} < 0,$$

the determinant condition (7) can hold only if

$$P_{12} < 0. \quad (8)$$

We now apply these conditions to the matrix

$$A = \begin{bmatrix} -k_1\rho & 1 \\ -2\gamma & 0 \end{bmatrix}, \quad (9)$$

where

$$\rho = |\text{PTF}_\beta(s)|, \quad \gamma = k_2\rho + d, \quad d = \text{sgn}(s)\delta. \quad (10)$$

Assume that

$$\rho \in [\rho_{\min}, \rho_{\max}], \quad |d| \leq \Delta, \quad (11)$$

where $\rho_{\min}$ and $\rho_{\max}$ are known positive constants.

Substituting

$$A_{11} = -k_1\rho, \quad A_{21} = -2(k_2\rho + d),$$

into (6) gives

$$-k_1P_{11}\rho - 2(k_2\rho + d)P_{12} < 0. \quad (12)$$

Equivalently,

$$(k_1P_{11} + 2k_2P_{12})\rho + 2dP_{12} > 0. \quad (13)$$

Since $P_{12} < 0$, the worst case with respect to d is obtained for $d = \Delta$. Hence, a sufficient condition ensuring (13) for all $\rho \in [\rho_{\min}, \rho_{\max}]$ and $|d| \leq \Delta$ is

$$(k_1P_{11} + 2k_2P_{12})\rho_{\min} + 2\Delta P_{12} > 0. \quad (14)$$

The determinant condition (7) can be written as

$$\Pi(\rho, d) > 0, \quad (15)$$

where

$$\begin{aligned} \Pi(\rho, d) &= 4P_{12}(-k_1P_{11}\rho - 2P_{12}(k_2\rho + d)) \\ &\quad - (P_{11} - k_1P_{12}\rho - 2P_{22}(k_2\rho + d))^2. \end{aligned} \quad (16)$$

Equivalently,

$$\Pi(\rho, d) = a_2\rho^2 + a_1\rho + a_0, \quad (17)$$

with

$$a_2 = -(k_1P_{12} + 2k_2P_{22})^2, \quad (18)$$

$$a_1 = -4P_{12}(k_1P_{11} + 2k_2P_{12}) + 2(P_{11} - 2dP_{22})(k_1P_{12} + 2k_2P_{22}), \quad (19)$$

and

$$a_0 = -8dP_{12}^2 - (P_{11} - 2dP_{22})^2. \quad (20)$$

Since

$$a_2 \leq 0, \quad (21)$$

the polynomial $\Pi(\rho, d)$ is concave with respect to ρ . Therefore, for a fixed admissible value of d , the condition

$$\Pi(\rho, d) > 0, \quad \forall \rho \in [\rho_{\min}, \rho_{\max}], \quad (22)$$

is guaranteed if and only if

$$\Pi(\rho_{\min}, d) > 0, \quad \Pi(\rho_{\max}, d) > 0. \quad (23)$$

Equivalently,

$$a_2\rho_{\min}^2 + a_1\rho_{\min} + a_0 > 0, \quad (24)$$

and

$$a_2\rho_{\max}^2 + a_1\rho_{\max} + a_0 > 0. \quad (25)$$

Since a_1 and a_0 depend on d , the inequalities (24) and (25) must be verified for all $|d| \leq \Delta$.

Consequently, a sufficient set of conditions ensuring

$$P > 0, \quad A^\top P + PA < 0, \quad (26)$$

for all $\rho \in [\rho_{\min}, \rho_{\max}]$ and $|d| \leq \Delta$, is given by

$$P_{11} > 0, \quad (27)$$

$$P_{11}P_{22} - P_{12}^2 > 0, \quad (28)$$

$$P_{12} < 0, \quad (29)$$

$$(k_1P_{11} + 2k_2P_{12})\rho_{\min} + 2\Delta P_{12} > 0, \quad (30)$$

$$a_2\rho_{\min}^2 + a_1\rho_{\min} + a_0 > 0, \quad \forall |d| \leq \Delta, \quad (31)$$

and

$$a_2\rho_{\max}^2 + a_1\rho_{\max} + a_0 > 0, \quad \forall |d| \leq \Delta. \quad (32)$$